Homework 2: Financial Data Analysis

FMA4200@CUHK(SZ)

Due on March 14, 2026 (11:59pm)

Submit an electronic version of your answer (report and codes files) to Blackboard (no late submission will be accepted).

1. Plot the Bitcoin price series (`bitcoin_data.csv`) as well as its log returns. Comment on the series independence and stationarity based on your visualizations.
2. Compute and plot the ACF at various lags for the Bitcoin log returns, absolute returns and squared returns. Comment on your findings regarding short- and long-range dependence or volatility clustering.
3. Apply the Augmented Dickey-Fuller (ADF) test to both the price and log return series. State the null and alternative hypotheses of the ADF test and use your computed results to determine whether each series is stationary or non-stationary.